Independent solve4 suffix-update review

Task T-1441. 3 November 2026.

Field update	Suffix = full subtraction under zero prefix
Word refinement	Canonical operands and p at most 2^32
Boundary controls	Wrong multiplier / premise / overflow
Fresh machine review	10 producer + 9 controls + 5 reviewer

One target-row update reviewed; full solve4 loop proof remains next. T-1441.

Ten producer declarations and nine producer controls passed fresh Lean replay, together with five reviewer declarations and 24 named axiom readbacks. The scoped verdict is validated: the finite-function model of one target-row suffix update matches the native arithmetic expressions and preserves the augmented equation under its explicit premises. Producer c59e9400a1 was published before replay.

Field argument. Full subtraction changes residual by minus the captured multiplier times the pivot residual. Zero pivot coefficients before col make suffix-only subtraction identical to full subtraction. Consequently equation equivalence needs the pivot row to satisfy the equation. Normalization to 1 independently makes the target pivot coefficient zero. The assumptions are satisfiable, including the mod 7 producer fixture; zero multiplier is covered.

Native correspondence. The actual bound Rust source captures z=mat[i][col] before the inner loop, iterates j from col through 4 and updates the right-hand side too. The model uses (a+p-b)%p and a*b%p. Canonical inputs and p<=2^32 bound multiplication below 2^64 and keep addition/subtraction intermediates valid, so both Rust checked arithmetic and UInt64 arithmetic have the intended value on this domain. The prime instance supplies the field; sub_decode itself does not need primality. Outputs remain canonical. The producer native-source excerpt is a concatenation of snippets, not a standalone compilable Rust file; actual bound source hashes were checked.

Independent attacks. A deliberately wrong finite fold recaptures the current pivot coefficient at every iteration; it yields [0,0,5,0,0] instead of [0,0,1,0,1] for the mod 7 fixture. A second countermodel keeps a normalized zero-prefix pivot but drops its equation, breaking preservation. Additional checked countermodels use a noncanonical subtraction operand and modulus 2^32+1 to expose overflow beyond the theorem domain. These supplement the producer prefix, RHS and normalization controls. A deliberately false RHS theorem was rejected with exit 1.

Verification and provenance. One actual six-command review run completed in 18.127512 seconds; all children reaped, 28,798 input file hashes unchanged. The 24 named axiom readbacks use only propext, Classical.choice and Quot.sound. The first wrapper failed before Python/Lean because OS stack hard limit was 10 MiB; a prospective amendment retained that stricter limit with Lean internal 16 MiB stack, 8 GiB address space, 3500 MiB heap, one worker and 295+5 second outer ceiling. A preparation path error before launch is retained separately. Local imported source and cached oleans were copied; upstream caches were hash-bound but not independently kernel-rechecked or rebuilt.

Review independence. Reviewer authored only the prior ordinary-fiber review, not these proofs or native implementation. Prior context and producer report access are disclosed. The residual derivation was recorded before new producer source access. This is non-author source-fidelity review, not a claim of blind derivation or a different algorithm.

Boundary and next construction. The suffix theorem assumes zero pivot prefix and normalization. Deriving those premises from pivot search, row swaps, modular inversion and prior-column induction is next. Actual nested array mutation, row alias exclusion, complete solve4 return correctness and counters remain distinct obligations. No N19 cohort, saved matrix run, cloud computation or performance measurement was rerun.